Sheet 13

Context values, locality and trajectories

1. Evaluate the six signed context terms and χ for the all-+1 assignment to the Peres–Mermin square, then for the assignment with only $v(A_{11}) = -1$. Which quantum context product does the first assignment fail? What extra independence assumption is needed when averaging assignments sampled in different experimental contexts?
2. A toy response model has $\lambda = \pm 1$ with equal weights, settings $x, y \in \{0, 1\}$, and deterministic outcomes $\alpha = \lambda$, $\beta = \lambda(-1)^{xy}$. Check PI, OI and averaged no-signalling. Compute its CHSH value and compare with the quantum upper value $2\sqrt{2}$ recalled in QM-I. This toy is not asserted to be a quantum or Bohmian realisation.
3. **A non-vacuous guiding flow (longer).** A freely spreading Gaussian is

$$\psi_t(x) = (2\pi\sigma^2)^{-1/4}(1 + it/\tau)^{-1/2} e^{-x^2/[4\sigma^2(1+it/\tau)]}, \qquad \tau = 2m\sigma^2/\hbar.$$

   Compute its velocity, solve $\dot{X}_t = v_t(X_t)$, $X_0(a) = a$, and verify transport of the Gaussian position density. Does any trajectory cross zero? Which global-existence difficulties excluded by this example still need the cited general theorem for other states?

> **First hint**
>
> Differentiate the logarithm of the packet with respect to position. Only its imaginary part contributes to the guiding velocity. Separate the real and imaginary parts of the complex width before solving the trajectory equation.

> **Further hint**
>
> The velocity is linear in position, so integrate the logarithmic change of a nonzero trajectory's absolute value and then handle the zero trajectory separately. For density transport, compare the Gaussian width and the Jacobian of the trajectory map. This example is nowhere zero; list which hypotheses that makes easy rather than generalising to packets with nodes.